1 Abstract

Classical *Schrödinger Bridge* (SB) problem seeks a path measure that minimally deviates from a given reference measure subject to prescribed initial and terminal marginals. This works investigate numerical approach to dynamic of interacting brownian particle with normal reflection term in low-dimensional setting ($d \leqslant 3$) on bounded domain $\mathcal{D} \subset \mathbb{R}^d$. To alleviate the difficulty due to nonlinearity incurred by non-local coupling, a nested fixed-point is adopted with Sinkhorn-type iterates. Further, fully semi-lagrangian discretization is used to propagate the solution of forward-backward partial differential equations (FB-PDEs).

2 Introduction

3 Related Works

Interest in spatially-constrained SB is largely under-developed, especially when also incorporating mean-field framework. This work is inspired by [4, 9]. First paper studies SB with reflected dynamic with Sinkhorn iterate where the time marching solution involve Neuman and Rudin boundary conditions, and show great performance in low-dimension. The latter generalized standard Sinkhorn to accept partial-integro-PDEs associated with mean-field SB resulting in nested version. Along this line [1] studies degenerate, sub-Riemannian dynamics, and use Sinkhorn simiarl to the former. Recent work [13] use Finite element discretization to solve the weak form of coupled PDEs for reflected SB. [8] derives forward-backward SDEs associated coupled FB-PDEs of reflected SB and extend divergence-based likelihood training in smooth bounded domain and shown to scale in high-dimension. Discrete-time framework also been explored with chance-constrants [15, 16] with mixture of Guassian marginal contraints. Alternative method such as introducing direct soft-penalty for particle escaping region exists as demonstrated in [12] using FB-SDE representation. There is a direct link between system of optimality of SB and MFG, therefore one can naturally adopt a scheme in MFG such as finite-difference to solve SB problem. [5–7] adopt a semi-lagrangian approach.

4 Problem Formulation

Let $\mathcal{D} \subset \mathbb{R}^d$ and boundary $\partial\mathcal{D}$, represented by smooth level-set $\chi(t,x)$ via $\chi < 0$ on $\mathcal{D}$ and $\chi = 0$ on $\partial\mathcal{D}$. Assuming $\nabla\chi \neq 0$ on $\partial\mathcal{D}$, the outward unit normal $\mathbf{n}(x)$ is obtained by $\mathbf{n} = \nabla\chi/\|\nabla\chi\|$. Denote by $\mathscr{P}(\mathcal{S})$ the space of probability measures on $\mathcal{S}$ endowed with the 2-Wasserstein metric $W_2(\cdot,\cdot)$ and having finite 2nd-moment. Let $\Omega :- \mathcal{C}(\mathcal{T};\mathbb{R}^d)$ where $\mathcal{T} :- [0,T]$ be canonical path space. Let $\mathbb{P}_t \in \mathscr{P}(\mathbb{R}^d)$ be t-time pushforward marginal of $\mathbb{P} \in \mathscr{P}(\Omega)$. For given fixed $\mathbb{P}$ and $\mu_{\text{init}} \in \mathscr{P}(\mathbb{R}^d)$, consider the dynamic of weakly-interacting brownian particle modeled by Mckean-Vlaskov reflected SDEs:

$$dX_t = -(\nabla E \star \mathbb{P}_t)(X_t) + \sigma\ dW_t - \mathbf{n}_t dV_t \qquad , X_0 \sim \mu_{\text{init}} \tag{1}$$

which induces the path law $\Gamma(\mathbb{P}) :- \text{Law}\left((X_t)_{t\in\mathcal{T}}\right)$, W is $\mathbb{R}^d$-valued Wiener on $(\Omega, \mathcal{F}, \mathbb{F} = (\mathcal{F}_t)_{t\in\mathcal{T}}, \mathbb{P}_0)$ Here E is potential, $(\nabla E \star \mu)(\cdot) :- \int_{\mathbb{R}^d} \nabla E(\cdot - y)\ \mu(dy)$. And V is a continuous, non-decreasing, $\mathbb{F}$-adapted bounded-variation process satisfying $V_0 = 0$, $\int_0^T \mathbf{1}_{\{X_s \notin \partial\mathcal{D}_s\}}\ dV_s = 0$, and $V_t = \int_0^t \mathbf{1}_{\{X_s \in \partial\mathcal{D}_s\}}\ dV_s$. For simplicity takes diffusion be $\sigma \equiv \sqrt{2\kappa}\mathbb{I}_d$ for $\kappa_0 > 0$. In [] shows that the rate function governing the large deviations for mean-field type dynamic in (1) is $H(\mathbb{P} \parallel \Gamma(\mathbb{P}))$. Unlike standard non-mean-field SB in that the referenc measure is no longer fixed and the convexity property is losed due to nonlinear dependency in its arguments of relative entropy []. Under appropriate hypotheses, the existence of $\mathbb{F}$-progressively measurable $\alpha(t,x)$ is guaranteed by Girsanov's, the relative entropy can be parametrized by process α:

$$H\big(\mathbb{P} \parallel \Gamma(\mathbb{P})\big) = \frac{1}{2}\mathbb{E}^{\mathbb{P}}\left[\int_0^T \|\alpha(t,X_s)\|^2\ ds\right] \tag{2}$$

and that X_t follows the controlled Mckean Vlasov SDE:

$$dX_t = b^\alpha(t, X_t, \mathbb{P}_t)\ dt + \sigma_t\ dW_t^{\mathbb{P}} - \mathbf{n}_t dV_t, \quad X_0 \sim \mu_{\text{init}}, \tag{3}$$

where $W^{\mathbb{P}}$ is a standard $(\mathbb{P}, \mathcal{F}_t)$-Brownian, $b^\alpha(t,x,\mu) :- \sigma_t\alpha_t - \nabla E \star \mu$. Further, under additional hypothesis, the perturbation is markovian feedback control. i.e. there exists $\Phi_t \in H_{-1}(\rho_t : t \in \mathcal{T})$ such that $\alpha_t = \Phi_t(X_t)$, $dt \times d\mathbb{P}$ a.s. (here assuming the control dimension is similar to state dimension). Following [], using transformed control $m := \rho\alpha \in \mathbb{R}^d$, rewrite (2) as SOC problem

$$\mathcal{J}^*_{\text{rmf-sb}} = \inf_{(\alpha,\rho)} \mathcal{J}_{\text{rmf-sb}}(m,\rho), \text{ where } \mathcal{J}_{\text{rmf-sb}}(m,\rho) :- \frac{1}{2}\int_0^T \int_{\mathbb{R}^d} \|m_t\|^2/\rho_t dx\ dt. \tag{4}$$

The law of sample path $(X_t)_{t\in\mathcal{T}}$ in (3) can be characterized as solving the martingale problem, i.e. under suitable assumption, for any $\psi \in \mathcal{C}^2(\mathcal{D})$, the process

$$M_t :- \psi(X_t) - \psi(X_0) - \int_0^t \mathcal{G}[\rho_s]\psi(X_s)ds \tag{5}$$

is $\mathbb{P}$-local martingale with respect to $\mathbb{F}$ where $\mathcal{G}$ is generator for (3) defined by $\mathcal{G}[\rho] :- \kappa\nabla\rho + \nabla\rho\cdot(\sigma\alpha - E\star\rho)$. It follows that [], the marginal flows $(\rho_t)_{t\in\mathcal{T}}$ is a weak solution of forward Kolmogorov equation:

$$\begin{aligned} \partial_t\rho + \nabla\cdot J = 0,\ (t,x)\in\mathcal{D},\\ J\cdot\mathbf{n} = 0,\ (t,x)\in\partial\mathcal{D} \end{aligned} \tag{6}$$

where probability flux is $J(m,\alpha) = \sqrt{2\kappa}m - \rho\nabla E\star\rho - \kappa\nabla\rho$, where here using $m :- \rho\alpha$. One can now characterize the 1st-order optimality condition associated with (4) with contraint (6) which yields a system of forward-backward PDEs. Here the system is adjacent to that appear in [], supplemented with additional Neuman and Robin boundary conditions. The proof given here is straightforward application of Lagrange mulitplier on Banach space.

4.1 Assumption

- The domain $\mathcal{D}\subset\mathbb{R}^d$ is sufficiently regular.
- Interacting term $E\in\mathcal{C}^2\left(\mathbb{R}^d;\mathbb{R}\right)$ is symmetric, i.e. $E(\cdot) = E(-\cdot)$ with $\sup_{z,v\in\mathbb{R}^d} v^\top\nabla^2E(z)v < +\infty$.
- The admissible control process $\Theta_\alpha = L^2$. And Θ_ρ be such that $\rho\in L^2\left(0,T;H^1\right)\cap\mathcal{C}\left(\mathcal{T};L^2\right)$ with $\rho\geqslant 0$, $\int_{\mathcal{D}}\rho = 1$, $\rho(0,\cdot)dx = \mu_{\text{init}}$, $\rho(T,\cdot)dx = \mu_{\text{targ}}$. And $\Theta_m = L^2\left(0,T;L^2\right)$.

The rest of this work focus solving the following coupled FB-PDEs which can be drived in straightforward manner by mean of Lagrange Multiplier on Banach space. The difference between (PDE-opt) and that which given by [2] is the extra boundary condition.

Proposition 1. *There exists $\Phi\in H^{-1}$such that the optimal control α^*,ρ^* are given by $\alpha^* = -\sqrt{2\kappa}\nabla\log\phi$ where $(\phi,\rho)\to(\varphi,\hat\varphi) :- (e^{-\phi},\rho\exp(E\star\rho/\kappa+\phi)$ solve the following FB-PDEs:*

$$\begin{cases}\partial_t\varphi = \left(\mathcal{A}^\rho + \mathcal{R}^\rho\right)[\varphi]\\ \partial_t\hat\varphi = -\left(\mathcal{A}^\rho+\mathcal{R}^\rho\right)[\hat\varphi]\end{cases},\ \text{on } (0,T)\times\mathcal{D},\quad \text{such that } \begin{cases}\nabla\varphi\cdot\mathbf{n} = \nabla\hat\varphi\cdot\mathbf{n} = 0,\ \text{on } (0,T)\times\partial\mathcal{D}\\ \mu_{init} = W[\rho]\varphi_0\hat\varphi_0,\ \text{on } \{0\}\times\mathcal{D}\\ \mu_{targ} = W[\rho]\varphi_T\hat\varphi_T,\ \text{on } \{T\}\times\mathcal{D}\end{cases} \tag{7}$$

where $\mathcal{A}^\rho[\eta] :- (\nabla E\star\rho)\cdot\nabla\eta - \kappa\Delta\eta$ is advection-diffusion operator and $\mathcal{R}^\rho[\eta] :- -\eta\nabla E\star(\rho\cdot\nabla\log\eta)$ is nonlocal source term, and $W[\rho](t,x) :- e^{-E\star\rho/\kappa}$.

5 Numerical Scheme

The idea is to constructive iterative scheme to find fixed point of marginal flow $\rho = (\rho_t)_{t\in\mathcal{T}}$ given by $F : \mathcal{C}\left(\mathcal{T};\mathscr{P}\left(\mathbb{R}^d\right)\right)\to\mathcal{C}\left(\mathcal{T};\mathscr{P}\left(\mathbb{R}^d\right)\right)$. And for fixed ρ^n at iteration $n = 1,...,N_{\text{iter}}$, the inner map for potential pairs $(\varphi^*,\hat\varphi^*) = G\left((\varphi^*,\hat\varphi^*);\rho^n\right)$. Define

$$F\left(\rho;\varphi^*,\hat\varphi^*\right) :- W[\rho]\varphi^*\hat\varphi^*\quad\text{and}\quad G(\cdot;\mu) :- M_T\circ S^\dagger_{0\to T}\circ M_0\circ S_{T\to 0}. \tag{8}$$

Here $S_{0\to T}$ and $S_{T\to 0}$ the forward and backward propagator operator of FB-PDEs in (PDE-opt) respectively. The operators M_0 and M_T are Sinkhorn updates that enforce marginal constrants, defined by

$$\hat\varphi_0 \xleftarrow{M_0(\varphi;\mu)} \frac{\mu_{\text{init}}}{W[\mu](x,0)\varphi(x,0)}\ ,\ \text{and}\quad \hat\varphi_T \xleftarrow{M_T(\varphi;\mu)} \frac{\mu_{\text{targ}}}{W[\mu](x,T)\varphi(x,T)}. \tag{9}$$

For fixed ρ^n (along with its convolution $\nabla E\star\rho^n$ and frozen log-gradient terms), the PDEs become linear Cauchy problems. And the time-varying operator $\mathcal{A}+\mathcal{R}$ (through density flow) admits unique evolution family $(P_{s,t})_{0\leqslant s\leqslant t\leqslant T}$, see also [3]. Over small $\Delta t = t_{k+1}-t_k$, the Strang Splitting regime for backward PDE for φ is defined as

$$\varphi(t_k)\approx S_{t_{k+1},t_k}[\varphi](t_{k+1},\cdot) :- (R_{t_{k+\Delta t/2},t_k}\circ U_{t_{k+1},t_k}\circ R_{t_{k+1},t_k+\Delta t/2})[\varphi](t_{k+1},\cdot) \tag{10}$$

where $R_{s,t}[\varphi] = \varphi\exp\int_s^t r_u du$ where $r\in L^\infty(\mathcal{D})$ since R is bounded multiplicative. For diffusion-advection subproblem, U has probabilistic representation [14] given by Markov propagator

$$U_{s,t}[\eta](x) :- \mathbb{E}^{\mathbb{P}}\left[\eta(t,X_t)|X_s = x\right] = \lim_{K\to\infty}\textstyle\sum_{1\leqslant m\leqslant K} w_m\eta\left(t,X^x_{s,t}(\omega_m)\right) \tag{11}$$

with appropriate weight positive, normalized w_m. Here X_t is chosen to be a solution to reference reflected-SDE in (1) with initial condition $X_s = x$ for fixed ρ^n, and. For finite K, a typical variance of random variable $\eta\left(t, X^x_{s,t}\right)$ in (11) has size t, and the by law of large number for finite K the estimate error is of order $O\left(\sqrt{T/K}\right)$. Let $S_{T\to 0} :- \prod_{j=N_T-1}^{0} S_{k,k+1}$. It follows that the that Strang Splitting satisfies 2nd-order truncation bound [3], for some $C \geqslant 0$:

$$\|P_{0,T}\varphi_T - S_{T\to 0}\varphi_T\|_{L^2} \leqslant C \cdot \Delta t^2 \|\varphi_T\|_{L^2} \tag{12}$$

Similarly, since forward PDE is formal L^2-adjoint of the backward one, so $S^{\dagger}_{t_k,t_{k+1}}$ is exactly given by reversing operation:

$$\hat{\varphi}(t_{k+1}) \approx S^{\dagger}_{t_k,t_{k+1}}[\hat{\varphi}](t_k,\cdot) = \left(R^{\dagger}_{t_{k+\Delta t/2},t_{k+1}} \circ U^{\dagger}_{t_k,t_{k+1}} \circ R^{\dagger}_{t_k,t_{k+\Delta t/2}}\right)[\hat{\varphi}](t_k,\ \cdot) \tag{13}$$

The useful property is that, upon spatial discretization, the discrete adjoint $S^{\dagger}$ can be obtained efficiently via matrix transposition of S, see also [10]. The below algorithm summarizes the scheme. Here for $S\left[\cdot;\rho^n,\varphi^n\right]$ indicates fixed evaluation point using ρ^n and φ^n.

Algorithm 1: Solver 1: Fixed-Point Iteration

for $n = 1, \ldots, N_{iter}$ **do**

1. $\varphi^k \leftarrow S_{T\to 0}\left[\varphi^{n-1}_T; \rho^{n-1}, \varphi^{n-1}\right]$ (backward propagate)
2. $\hat{\varphi}_0 \leftarrow M_0\left[\varphi^n; \rho^{n-1}\right]$ (initial marginal scale)
3. $\hat{\varphi}^k \leftarrow S^{\dagger}_{0\to T}\left[\hat{\varphi}_0; \rho^n, \hat{\varphi}^{n-1}\right]$ (forward propagate)
4. $\varphi^{\text{next}}_T \leftarrow M_T\left[\hat{\varphi}^n; \rho^{n-1}\right]$ (terminal marginal scale)
5. $\rho^n \leftarrow F\left[\rho^{n-1}; \varphi^n, \hat{\varphi}^n\right]$ (density update)

The main difference between Solver 1 [11] is that it only comprise one loop iteration without nested structure. The motivation behind this construction the ability to adopt a type of fictitous play in the MFG setting (or a Picard-iterate type), while able to adopt the standard Sinkhorn scaling method which only applicable to narrow set of PDEs structure.

5.1 Semi-Lagrangian Discretization

What left now is to discretize in space via SL. It is known that [5] SL scheme is monotone, consistent and stable for HJB, contrary to finite difference, therefore the transformed variable converges to exact true field under appropriate conditions. Let $\mathcal{D}_\Delta = \{x_i\}_{1\leqslant i\leqslant N_x} \subset \mathcal{D}$ a nodes of a simplicial triangulation over the polyhedral domain $\mathcal{D}$, and time subdivision $\mathcal{T}_\Delta = \{t_k = k\Delta t\}_{0\leqslant k\leqslant N_t}$. At t_k, the continuous field $\eta \in L^2(\mathcal{D})$ is approximated by nodal values $\eta_k \in \mathbb{R}^{N_x}$. Its continuous reconstruction is obtained with P1 finite element interpolator $\mathcal{I} : L^2(\mathcal{D}_\Delta) \to \mathcal{C}_b(\mathcal{D})$ defined by $\mathcal{I}[\eta](\cdot) :- \beta(\cdot)^\top \eta$ where $\beta = (\beta_1, \ldots \beta_{N_x})^\top$ is vector of standard piecewise linear Lagrange basis with $\beta_i(x_j) = \delta_{ij}$ and $\mathbf{1}^\top \beta = 1$. This interpolation introduces a spatial discretization error strictly bounded by $O(\Delta x^2)$. Following [7], choose $\beta_i(a) = \max\{0, 1 - |a - x_i|/\Delta x\}$ which has support on $[x_i - \Delta x, x_i + \Delta x]$. The approximate of $U_{t_k,t_{k+1}}[\eta](x_i)$ is given by stochastic 3-branches stencil. For each node $x_{k,i} \in \mathcal{D}_\Delta$, the advective transport under reference drift $\mathbf{b}_{k,i} \in \mathbb{R}^d$ with diffusion $2\kappa\mathbb{I}_d$ is approximated by (unconstrained) backward characteristic path

$$y^s_{k,i,l}(a) = x_i - \Delta t b_{k,i} + e_l s\sqrt{6\kappa d\Delta t}, \qquad s \in \{0, \pm 1\},\ l \in \{1, ..., d\}. \tag{14}$$

where e_l is the canonical basis. The Neuman reflection boundary can be enforced to characteristic that exit the domain, following [5] for special case of normal reflection. Specifically, recall that $\chi < 0 \iff x \notin \mathcal{D}$ is chosen to be sign-distance function, the Skorokhod map reads

$$\hat{y}^s_{k,\mathbf{i},\ell} := \begin{cases} y^s_{k,\mathbf{i},\ell} & \text{if } y^s_{k,\mathbf{i},\ell} \in \mathcal{D}, \\ y^s_{k,\mathbf{i},\ell} - 2\chi\left(y^s_{k,i,l}\right)\mathbf{n}\left(\chi\left(y^s_{k,i,l}\right)\right) & \text{if } y^s_{k,\mathbf{i},\ell} \notin \mathcal{D}. \end{cases} \tag{15}$$

where $\mathbf{n}(\chi(a)) = \nabla\chi(a)/\left\|\nabla\chi(a)\right\|_2$. Using (11) and 3 stencil weights, the fully discrete SL approximate at x_i is given by $U_{t_k,t_{k+1}}[\eta](x_i) \approx \sum_{1\leqslant j\leqslant N_x}(\mathbf{\Pi}_k)_{i,j}\eta_j$, where $\mathbf{\Pi}_k \in \mathbb{R}^{N_x\times N_x}$ is given by

$$(\mathbf{\Pi}_k)_{a,b} :- \tfrac{2}{3}\beta_j\left(\hat{y}^0_{k,i,1}\right) + \tfrac{1}{6d}\sum_{\ell=1}^{d}\beta_j\left(\hat{y}^+_{k,i,\ell}\right) + \beta_j\left(\hat{y}^-_{k,i,\ell}\right) \tag{16}$$

This choice is has 2nd-order accurachy: $\mathbb{E}[\Delta Y_i] = -\Delta t b_k(x_i)$ and $\mathrm{Cov}(\Delta Y_i) = 2\kappa\Delta t\mathbb{I}_d$,which is congruent with $\mathbb{E}[dX_t]$ and $\mathrm{Cov}\left(\sqrt{2\kappa}dW_t\right)$, respectively. For probabilistic interpretation in term of controlled-markov process

of SL scheme, see [5]. Define

$$\begin{aligned} \mathbf{V}_k &:= \textstyle\sum_{1\leqslant j\leqslant N_x} E(x_i - x_j)\rho_{k,j}m_j \quad , \quad \mathbf{W}_k := \operatorname{diag}\exp(-\mathbf{V}_k/\kappa) \\ \mathbf{R}_{n,\pm 1/2} &= \operatorname{diag}\exp\left(\pm\tfrac{\Delta t}{2}\boldsymbol{r}_k^{n-1}\right) \quad , \quad \mathbf{r}_k^n = -\mathbf{b}_k^\top \nabla\log(\varphi_k/\rho_k) \end{aligned} \tag{17}$$

Note also that the discrete L^2-adjoint for forward $U_{t_k,t_{k+1}}$ is simply transpose given by $\mathbf{\Pi}_k^\dagger = M_{\text{lump}}^{-1}\mathbf{\Pi}_k^\top M_{\text{lump}}$ where $M_{\text{lump}} = \operatorname{diag}(m_1, ..., m_{N_x}),\ m_i = \int_{\mathcal{D}} \beta_i(a)da$.

Algorithm 2: Algorithm 1: (Matrix-form for Solver 1)

Input: $\mu_{\text{init}}, \mu_{\text{targ}} \in \mathscr{P}(\mathcal{D}_\Delta)$, $\Delta t = T/N_T$, $\theta \in (0,1]$, threshold $\epsilon > 0$
Initialize: $\varphi_k^0, \hat{\varphi}_k^0 \leftarrow \mathbb{1}_{N_x}$ and $\rho^0 \leftarrow \mathbb{1}_{N_x}/\text{Vol}_d(\mathcal{D})$, for $k = 0, ..., N_T$;
for *each* $n = 1, .., N_{iter}$ **do**
  Compute $b_k^{n-1} \leftarrow -(\nabla E \star \rho)_k$, $W_k^{n-1} \leftarrow \operatorname{diag}\exp\left(-\mathbf{V}_k^{n-1}/\kappa\right)$, for $k = 0, ..., N_T$;
  for *each* $k = N_t - 1, N_t - 2, ..., 0$ **do**
    $\boldsymbol{\varphi}_k^n = \mathbf{R}_{k,+1/2}^{n-1}\mathbf{\Pi}_k\mathbf{R}_{k+1,-1/2}^{n-1}\boldsymbol{\varphi}_{k+1}^n$;
  Set $\hat{\boldsymbol{\varphi}}_0^n = \boldsymbol{\mu}_{\text{init}} \oslash \left(\mathbf{W}_0^{n-1} \odot \boldsymbol{\varphi}_0^n\right)$;
  for *each* $k = 0, ..., N_T - 1$ **do**
    $\hat{\varphi}_{k+1}^n = \left(\mathbf{R}_{k+1,-1/2}^{n-1}\right)^{-1}\mathbf{\Pi}_k^\dagger\left(\mathbf{R}_{k,+1/2}^{n-1}\right)^{-1}\hat{\boldsymbol{\varphi}}_k^n$;
  Set $\boldsymbol{\varphi}_{N_T}^n = \boldsymbol{\mu}_{\text{targ}} \oslash \left(\mathbf{W}_{N_T}^{n-1} \odot \hat{\boldsymbol{\varphi}}_{N_T}^n\right)$;
  Set $\boldsymbol{\rho}_k^{1/2} = \mathbf{W}_k^{n-1} \odot \boldsymbol{\varphi}_k^n \odot \hat{\varphi}_k^n$ and then normalize $\rho^{1/2} \leftarrow \frac{\boldsymbol{\rho}_k^{1/2}}{\mathbf{1}^T M_{\text{lump}}\boldsymbol{\rho}_k^{1/2}}$, for $j = 0, ..., N_T - 1$;
  Update $\boldsymbol{\rho}_k^n = (1-\theta)\boldsymbol{\rho}_k^{n-1} + \theta\boldsymbol{\rho}_k^{1/2}$, for $j = 0, ..., N_T - 1$;
  if $\max_{1\leqslant k\leqslant N_{T-1}} \left\|\rho_k^n - \rho_k^{n-1}\right\|_\infty \leqslant \epsilon$ **then**
    break;

References

[1] Daniel Owusu Adu. Reflected Schrodinger Bridge Problem over Sub-Riemannian Manifold, 2026.

[2] Julio Backhoff-Veraguas et al. The Mean Field Schrödinger Problem: Ergodic Behavior, Entropy Estimates and Functional Inequalities. *Probability Theory and Related Fields*, 2020.

[3] András Bátkai, Petra Csomós, Bálint Farkas, and Gregor Nickel. Operator splitting for non-autonomous evolution equations. *Journal of Functional Analysis*, 260(7):2163–2190, 2011.

[4] Kenneth F. Caluya and Abhishek Halder. Reflected Schrödinger Bridge: Density Control with Path Constraints. In *American Control Conference*, 2020.

[5] Elisa Calzola, Elisabetta Carlini, Xavier Dupuis, and Francisco J. Silva. A semi-Lagrangian scheme for Hamilton-Jacobi-Bellman equations with oblique derivatives boundary conditions. *Numerische Mathematik*, 153(1):49–84, 2023.

[6] Elisabetta Carlini and Valentina Coscetti. A semi-Lagrangian scheme for first-order mean field games based on monotone operators. *IMA Journal of Numerical Analysis*, 2026.

[7] Elisabetta Carlini and Francisco J. Silva. Semi-Lagrangian schemes for mean field game models. In *2013 IEEE 52nd Annual Conference on Decision and Control (CDC)*, pages 3115–3120, 2013.

[8] Wei Deng, Yu Chen, Nicole Tianjiao Yang, Hengrong Du, Qi Feng, and Ricky T. Q. Chen. Reflected Schrödinger Bridge for Constrained Generative Modeling. arXiv preprint, 2024.

[9] Asmaa Eldesoukey, Yongxin Chen, and Abhishek Halder. A Generalized Sinkhorn Algorithm for Mean-Field Schrödinger Bridge. *IEEE Control Systems Letters*, 2026.

[10] Asmaa Eldesoukey, Md Zulfiqur Haider, Italo Napolitano, Yongxin Chen, and Abhishek Halder. Schrödinger Bridges over Kinetic Swarming Models, 2026.

[11] Asmaa Eldesoukey, Md Zulfiqur Haider, Italo Napolitano, Yongxin Chen, and Abhishek Halder. Second-Order Mean-Field Schrödinger Bridges with Partial State Information, 2026.

[12] Guan-Horng Liu et al. Deep Generalized Schrödinger Bridge, 2022.

[13] Wenxin Liu and Dante Kalise. A PDE-constrained Optimization Approach to Optimal Trajectory Planning under Uncertainty via Reflected Schrödinger Bridges. In *2026 European Control Conference (ECC)*, pages 1746–1751. IEEE, 2026.

[14] Grigori N. Milstein and Michael V. Tretyakov. Numerical algorithms for semilinear parabolic equations with small parameter based on approximation of stochastic equations. *Mathematics of Computation*, 69 (229):237–267, 2000.

[15] George Rapakoulias and Panagiotis Tsiotras. Discrete-time Optimal Covariance Steering via Semidefinite Programming, 2023.

[16] George Rapakoulias, Ali Reza Pedram, and Panagiotis Tsiotras. Steering Large Agent Populations Using Mean-Field Schrödinger Bridges With Gaussian Mixture Models. *IEEE Control Syst. Lett.*, 9:1760–1765, 2025.